

REPORT

ARTIFICIAL INTELLIGENCE (CSA1717)

Assessment Tool 3 – Resolution Algorithm Implementation

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Course Code: CSA1717

Course Name: Artificial Intelligence

Assessment Title: Resolution Algorithm Implementation

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RESOLUTION ALGORITHM IMPLEMENTATION

Introduction

Artificial Intelligence involves developing systems that can reason, learn, and make decisions based on available information. Logical reasoning is one of the fundamental techniques used by AI systems to derive new information from existing facts.

The Resolution Algorithm is a rule of inference commonly used in automated theorem proving. It works on logical clauses and provides a systematic method for proving statements. The algorithm is particularly useful because different logical expressions can be converted into a common CNF representation before resolution is applied.

In resolution-based proof, the statement that needs to be proved is negated. The negated statement is combined with the knowledge base. Resolution is then repeatedly performed. If the process produces an empty clause, a contradiction has been obtained. This contradiction proves that the original statement must be true.

This practical implementation applies the concept to five simple scenarios. Each scenario contains facts, rules, and a goal. The resolution process is carried out step by step to demonstrate logical inference.

Objectives

The main objectives of this practical are:

1. To understand the concept of propositional logic.
2. To represent real-world facts using logical propositions.
3. To convert logical implications into CNF.
4. To understand the concept of resolution.
5. To negate the goal before applying resolution.
6. To derive new clauses using complementary literals.
7. To obtain the empty clause as proof of contradiction.
8. To determine whether a given goal logically follows from a knowledge base.
9. To apply automated reasoning to simple real-world situations.

10. To understand the role of logical inference in Artificial Intelligence.

Theory

Propositional Logic

Propositional logic represents statements using propositions that can have either a **True** or **False** value.

For example:

- R = It is raining.
- W = The ground is wet.

A rule such as "If it rains, the ground becomes wet" can be represented as:

$$R \rightarrow W$$

Conjunctive Normal Form

Resolution requires logical expressions to be represented in **Conjunctive Normal Form (CNF)**.

An implication:

$$P \rightarrow Q$$

is converted into:

$$\neg P \vee Q$$

For example:

$$R \rightarrow W$$

becomes:

$$\neg R \vee W$$

Each expression separated by OR represents a clause.

Resolution Rule

The basic resolution rule is:

$$(P \vee Q), (\neg P \vee R) \Rightarrow (Q \vee R)$$

The complementary literals P and $\neg P$ are eliminated, producing a new clause.

6.4 Empty Clause

The **empty clause** ($\square$) represents a contradiction.

When the empty clause is derived after adding the negation of the goal, it proves that the original goal follows from the knowledge base.

Methodology

The following general methodology is used for every problem:

1. Identify the facts and rules.
2. Assign propositional symbols to the statements.
3. Convert the rules into propositional logic.
4. Convert implications into CNF.
5. Write the resulting clauses.
6. Identify the required goal.
7. Negate the goal.
8. Add the negated goal to the clause set.
9. Select clauses containing complementary literals.
10. Apply the Resolution Rule.
11. Continue until the empty clause is derived.
12. State the final conclusion.

Problem 1 – Rain and Wet Ground

Knowledge Base

- If it rains, the ground becomes wet.
- It is raining.

Propositions

- R = It is raining.
- W = Ground is wet.

CNF

$$R \rightarrow W$$

becomes:

$$\neg R \vee W$$

Fact:

R

Negated goal:

$\neg W$

Resolution

From:

$\neg R \vee W$

and

$\neg W$

we obtain:

$\neg R$

Then:

R

and

$\neg R$

produce:

□

Result

The empty clause is obtained.

Therefore, the ground is wet.

Problem 2 – Student Assignment Submission

Knowledge Base

- If a student submits the assignment, the student receives internal marks.
- Rahul submitted the assignment.

Propositions

- S = Rahul submitted the assignment.

- M = Rahul receives internal marks.

CNF

Rule:

$$S \rightarrow M$$

becomes:

$$\neg S \vee M$$

Fact:

$$S$$

Negated goal:

$$\neg M$$

Resolution

Resolve:

$$\neg S \vee M$$

with:

$$\neg M$$

to obtain:

$$\neg S$$

Then resolve:

$$S$$

with:

$$\neg S$$

to obtain:

□

Result

Therefore, Rahul receives internal marks.

Problem 3 – Library Membership

Knowledge Base

- If a person is a library member, the person can borrow books.
- Priya is a library member.

Propositions

- L = Priya is a library member.
- B = Priya can borrow books.

CNF

Rule:

$$L \rightarrow B$$

becomes:

$$\neg L \vee B$$

Fact:

$$L$$

Negated goal:

$$\neg B$$

Resolution

Resolve:

$$\neg L \vee B$$

and:

$$\neg B$$

to obtain:

$$\neg L$$

Then resolve:

$$L$$

and:

$$\neg L$$

to obtain:

□

Result

Therefore, Priya can borrow books.

Problem 4 – Placement Eligibility

Knowledge Base

- If a student clears the aptitude test, the student is eligible for placement.
- Arun cleared the aptitude test.

Propositions

- A = Arun cleared the aptitude test.
- P = Arun is eligible for placement.

CNF

Rule:

$$A \rightarrow P$$

becomes:

$$\neg A \vee P$$

Fact:

$$A$$

Negated goal:

$$\neg P$$

Resolution

Resolve:

$$\neg A \vee P$$

and:

$$\neg P$$

to obtain:

$\neg A$

Then resolve:

A

and:

$\neg A$

to obtain:

□

Result

Therefore, Arun is eligible for placement.

Problem 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, the user is authenticated.
2. If a user is authenticated, the user is granted access.
3. The user entered the correct password.

Propositions

- C = Correct password entered.
- A = User is authenticated.
- G = User is granted access.

CNF

First rule:

$C \rightarrow A$

becomes:

$\neg C \vee A$

Second rule:

$A \rightarrow G$

becomes:

$\neg A \vee G$

Fact:

C

Negated goal:

$\neg G$

Resolution

First:

$\neg A \vee G$

and:

$\neg G$

produce:

$\neg A$

Next:

$\neg C \vee A$

and:

$\neg A$

produce:

$\neg C$

Finally:

C

and:

$\neg C$

produce:

□

Result

Therefore, the user is granted access.

Results

Problem	Goal	Empty Clause Obtained	Result
Rain and Wet Ground	Ground is wet	Yes	Proved
Student Assignment	Rahul receives internal marks	Yes	Proved
Library Membership	Priya can borrow books	Yes	Proved
Placement Eligibility	Arun is eligible for placement	Yes	Proved
Access Control	User is granted access	Yes	Proved

The resolution process successfully proves the goal in all five cases.

Advantages

1. Provides a systematic method for logical inference.
2. Useful for automated theorem proving.
3. Can handle multiple logical clauses.
4. Reduces logical reasoning to a standard resolution process.
5. Clearly establishes whether a conclusion follows from given facts.
6. Useful in Artificial Intelligence and expert systems.
7. Helps understand automated reasoning techniques.
8. Provides a formal basis for proving logical statements.

Limitations

1. Resolution can become computationally expensive for large knowledge bases.
2. CNF conversion may increase the number of clauses.
3. It is mainly demonstrated here using propositional logic.

4. Large sets of clauses may produce many unnecessary intermediate clauses.
5. The method requires careful representation of facts and rules.

Conclusion

The Resolution Algorithm was successfully applied to five propositional logic problems. Each problem was represented using logical propositions, converted into CNF, and solved by negating the goal and applying resolution.

For every problem, the resolution process generated the empty clause ($\square$). This demonstrates that the negated goal creates a contradiction with the knowledge base. Therefore, the original goals are logically entailed by their respective knowledge bases.

The practical provides a clear understanding of how propositional logic, CNF conversion, contradiction-based reasoning, and resolution can be used to perform automated inference. These concepts form an important foundation for more advanced Artificial Intelligence reasoning systems.

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